

Newsletter (16) - Dual Use in Defense

(1) Introduction

In this week's version of our newsletter, we examine how dual use innovation is redefining Europe's defense landscape. Once a niche concept, the blending of civilian and military technologies has now become the dominant model for startups, investors, and governments alike. From drones to AI-driven software, civilian breakthroughs are being adapted for the battlefield at unprecedented speed, reshaping procurement and investment logic across the continent. We will explore why dual use has emerged as Europe's strongest lever for agility, its applications, how it creates spillover benefits beyond defense, and what challenges remain for Europe to secure sovereignty of its most critical technologies.

(2) Explanation of Dual Use

The conflict in Ukraine has changed how wars are fought. Heavy armor and fighter aircraft are no longer the decisive factors in modern warfare. Instead, affordable and widely available technologies from civilian markets have become essential.

Technology has emerged as the great equalizer. Low-cost, adaptable equipment, such as \$1,000 drones capable of disabling \$10 million armored vehicles, prove that innovation and speed can outweigh traditional military power. Dual use technology, where civilian innovations are adapted to defense, is driving this transformation. It marks a shift where progress in civilian technology directly shapes military strength.

For Europe, multi-purpose innovation has become essential for survival. Traditional defense models based around centralized procurement, state-led R&D and reliance on large defense contractors are too slow and costly, while civilian technology advances at a far quicker pace. Since 2022, more than 230 defense tech start-ups have emerged and funding has reached \$1.5 billion in 2025, the highest level on record. Europe's ability to remain secure and independent now hinges on its ability to translate startup agility into military capability.

(3) Applications of Dual Use in the Industry

The split in Europe's defense sector is clear. Pure defense firms remain trapped in the old cycle with decade-long procurement processes, revenue streams almost entirely state-reliant and limited room for maneuver once contracts are signed. Their dependence on slow-moving bureaucracies makes them unattractive to venture investors, who see little certainty and almost no scalability. By contrast, dual use companies have broken out of this trap. A drone designed for agriculture or mining can be adapted into an Intelligence, Surveillance, and Reconnaissance (ISR) platform for the battlefield; AI built for logistics can be turned into real-time targeting software. This model creates multiple exit opportunities in civil markets, defense primes or direct government acquisition, dramatically reducing investor risk.

The proof is visible across Europe: Quantum Systems, one of Europe's fastest growing defense firms, initially targeted commercial sectors including agriculture and infrastructure with its drone technology, before evolving into a full-fledged dual use player supplying ISR to military and government clients. Likewise, Helsing's €600M raise in 2025 shows how software-first dual use firms can attract capital at unprecedented scale, far beyond what pure defense startups could secure. This is not an exception, as according to Dealroom, 85% of European defense-tech startups now follow a dual use model straddling civil and military markets.

By contrast, US firms like Anduril and a few European players such as Stark remain defense-only, betting on becoming the next generation of primes rather than pursuing dual use scalability. Ultimately, the attraction is simple: it shortens time-to-mission, diversifies revenues, reduces capital intensity and creates models that investors can actually back.

(4) Dual Use Synergies Effect

Historically, civilian needs were met by technology evolved from military innovations, notable examples being GPS, radar and touchscreens. Today, the direction is reversed. Civilian breakthroughs in AI, robotics and drones set the pace, with defense racing to adapt them. Likewise, from a civilian perspective, speed of development is imperative. Innovation from the commercial ecosystem is far more prevalent owing to better funding and fewer constraints. Tech firms invest 10-70% of revenue in R&D, compared to just 1-4% for defense primes. This is why dual use startups outpace contractors. The agile, ready-to-pivot mindset of a start-up leads to innovation and resilience and incorporates real-world feedback into the design cycle. As a result, dual use not only accelerates defense, but also fosters mutual enrichment.

The unique upfront funding model of the defense industry highlights the need for dual use synergies: while traditional contractors struggle to deliver results on time and within budget, commercial start-ups only generate revenue after launching a marketable product. Civilian demand drives costs down, making military-grade systems affordable and adaptable. Conversely, military technology requires civil applications to remain commercially viable through cyclical defense needs. Dual use promotes cost efficiency and economies of scale, moreover, reducing risk for companies and investors.

(5) Europe's Challenge in capital and control

In recent history, Europe has been dependent on the US for both capital resources and technology. As of 2025 YTD, 45% of funding for European defense-tech startups comes from US investors, placing domestic IP, data, and export rights under foreign control. On the investment side, US firms have captured 85% of the NATO defense-tech VC market since 2019, thanks to deeper domestic capital pools and faster contracting, Dealroom reports. After years of underperformance, Europe has finally regained momentum in funding and start-up activity, yet it still hesitates to scale domestically as adversaries accelerate in face of rising tension in the Ukraine conflict. Dual use models set a new trend in the markets, putting ownership and control front and center.

Europe can consolidate its recent advances to create a self-sustaining defense ecosystem. Establishing a leading role for European capital in funding rounds will ensure that IP and data stay domiciled in Europe and diminish dependence on overseas investors. Concurrently, forming a unified, multi-year procurement framework among member states will strengthen long-term coordination and improve scalability. Coordinated orders provide companies with volume certainty, enabling industrial scale with lower unit costs.

(6) Conclusion

Dual use has emerged as Europe's most powerful lever for defense innovation. By collapsing the boundary between civilian and military markets, it accelerates development cycles, lowers capital intensity and creates scalable models that attract VC at record levels.

Yet the model alone is not enough. Without European capital in the lead, faster procurement and secure access to critical supply chains, much of this momentum risks being captured abroad. The challenge now is to turn Europe's dual use success stories into sovereign capabilities that remain under European control. The opportunity is clear: dual use is not optional; it is Europe's path to unity and sovereignty.